Designing an open-sourced alternative to Meta's sEMG-RD and fine-tuning on associated deep learning algorithms

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Introduction

Meta recently published three papers introducing a wireless, wrist-wearable surface electromyography (sEMG) device, several datasets collected using this device, and associated decoding algorithms for the tasks explored in these datasets. Their most notable contributions include accurate hand pose estimation and character classification during keyboard typing, as well as recognition of other hand-based activities, such as handwriting and swipe gestures [1,2,3]. However, because the datasets were collected using Meta's proprietary hardware, researchers, clinicians, and developers cannot leverage these advancements for inference without purchasing the device, and even then, usage is restricted to Meta's terms. In this paper, we present a closely matched open source alternative to their sEMG-RD system, along with a data collection and fine-tuning pipeline designed to enable broader adoption and reproducibility within the research community.

Background

Electromyography (EMG)

Electromyography measures the electrical activity of muscle tissue. More formally, "Electromyography (EMG) is an experimental technique concerned with the development, recording, and analysis of myoelectric signals. Myoelectric signals are formed by physiological variations in the state of the muscle fiber membranes" [4]. EMG signals can also be understood as the spatiotemporal summation of individual muscle fiber action potentials, providing an indirect measure of neural input to a muscle [5, p. 17]. The charges associated with muscle action potentials are on the order of nanocoulombs (nC), and the electric potential differences detected are typically in the microvolt to millivolt range [6]. Although values vary depending on muscle size, contraction type, and methodological differences, peak-to-peak amplitudes of up to 5 mV can be recorded during maximal isometric contractions using surface EMG [6]. Recording such small signals requires specialized instrumentation [7]. EMG can be measured intramuscularly, using a needle electrode inserted into the muscle, or noninvasively from the surface using

electrodes. It is primarily used for diagnosis, neuroprosthetics, human-computer interaction, and experimental research.

Surface Electromyography

Surface electromyography (sEMG) is a non-invasive technique for recording muscle activity using electrodes placed on the skin [8]. Compared to intramuscular EMG, sEMG is easier to apply and remove, making it more practical for repeated or user-operated measurements in clinical and research settings. The typical design of an sEMG acquisition system includes electrodes, differential preamplification, a high-pass filter, gain, detector, and a low-pass filter [5, p. 56]. The analog signals collected with this pipeline are then digitized by an Analog-to-Digital Converter (ADC) and sent to a computer or a microcontroller [9]. Key design considerations for an sEMG acquisition system include appropriate electrode placement and configuration (e.g., monopolar or bipolar), ensuring low electrode–skin impedance relative to the input impedance of the amplifier or ADC, and avoiding signal saturation caused by DC offsets or motion artifacts. Additionally, the system should incorporate proper anti-aliasing filters, exhibit a high common-mode rejection ratio (CMRR), maintain low input-referred noise, and provide sufficient amplification and resolution to accurately capture the low-amplitude EMG signals.

Electrodes

Surface electrodes can be broadly categorized into wet and dry types. Wet electrodes, typically made of silver/silver chloride (Ag/AgCl), use a conductive gel to reduce skin impedance and enhance signal quality. They are widely used in clinical environments due to their stable and reliable recordings. However, wet electrodes require skin preparation, may cause irritation with prolonged use, and can dry out over time, limiting their suitability for long-term or wearable applications [10,11].

In contrast, dry electrodes do not require conductive gel and are often made of metal, conductive fabric, or polymer-based materials [11]. They are more convenient for wearable or long-duration use and reduce preparation time, but generally have higher impedance and may be more susceptible to motion artifacts [10]. Advances in material science, such as those presented by Kaveh et al. [12], have improved the performance of dry electrodes, making them increasingly viable for consumer-grade and portable EMG systems. Their electrodes achieve “an average electrode-skin impedance of 66.7 kΩ at 50 Hz and DC offset of −20 mV without any hydrogel, which, when area normalized, is within the range achieved by wet electrodes

without skin abrasion." [12]. Further details on the system requirements for sEMG acquisition will be discussed in the later sections.

Meta's recent contributions

sEMG-RD

Meta recently introduced the sEMG-RD, a wrist-worn surface electromyography (sEMG) device featuring 16 channels [1,2,3]. These channels are implemented using 48 gold-plated brass pogo-pin style electrodes, aligned along the proximal-distal axis of the forearm. The electrodes are arranged in a bipolar configuration, with the remaining electrodes serving as ground or shielding. Each electrode has a diameter of 6.5 mm, and the center-to-center spacing between paired sensing electrodes is 20 mm. All 16 channels sample data at 2 kHz with 12-bit resolution, and the data is transmitted via Bluetooth to a host computer. The maximum signal amplitude is 6.6 mV, and signals are band-pass filtered at the analog front end between 20 Hz and 850 Hz (3 dB corner frequencies). The device exhibits an input-referred noise of 2 μVrms and can detect putative motor unit action potentials around the wrist under low-movement conditions for up to four hours. The analog sensors have a nominal gain of 190×, and the ADC has a full-scale range of 2.5 V, offering a dynamic range of approximately 65.5 dB. Although an IMU is present onboard, it was not utilized in the experiments discussed here. The input impedance of the analog front end is very high, approximately 10 pF in parallel with 10 TΩ. This device was utilized in data collection for three papers [1,2,3].

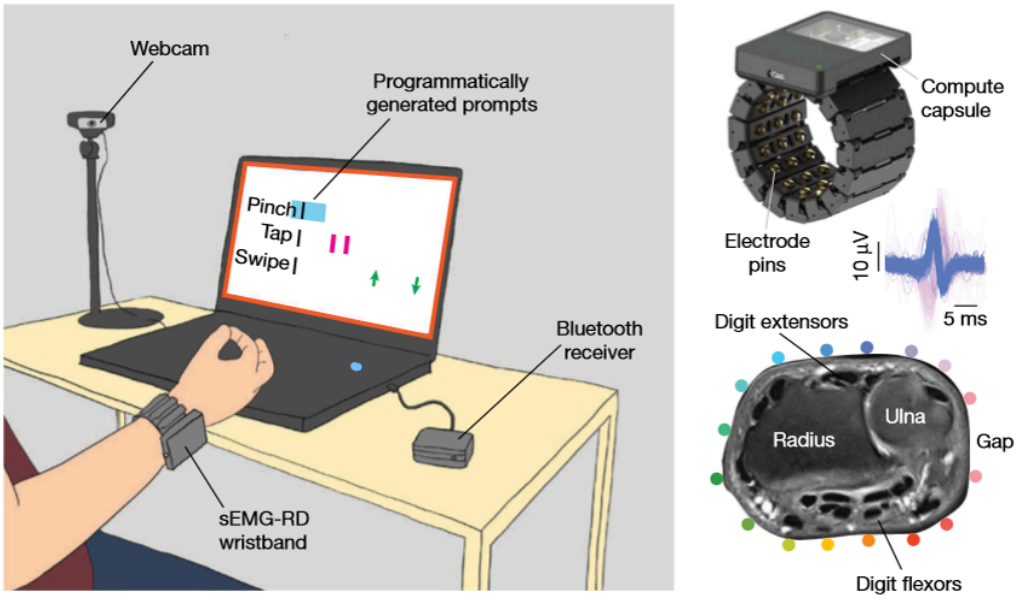


Figure 1: sEMG-RD resign and its placement on the wrist, adapted from [3]

As shown in Figure 1, sEMG-RD places its electrodes circumferentially around the wrist, with electrodes of the same channel placed along the proximal–distal axis over the wrist region, rather than over the forearm muscle bellies. This placement is supported by anatomical and experimental evidence showing that wrist-level sEMG captures muscle activity directly involved in fine finger control. In particular, Botros et al. report that "there is a unique group of muscles that are only present at the wrist, near the distal end of the forearm, proximal to the ulnar styloid process, and control fine finger and thumb movements. These muscles are the abductor pollicis longus (APL), extensor pollicis brevis (EPB), and extensor pollicis longus (EPL). Additionally, the extensor indices (EI) muscle extends the index finger, along with the ED and EDM. On the anterior side of the wrist, there is the flexor policis longus (FPL) muscle that flexes the thumb and the FDS muscle that flexes the medial four fingers and assists with hand flexion at the wrist. Additionally, there is the pronator quadratus (PQ) muscle, which helps in pronating the whole forearm" [32]. Systematic comparisons between wrist and forearm recordings further demonstrate that wrist-based sEMG yields significantly higher signal quality and classification performance for single-finger and multifinger gestures, while maintaining comparable performance for wrist gestures [32]. Taken together, these results support wrist-based electrode placement as an anatomically grounded and experimentally validated choice for decoding fine hand and finger movements.

emg2pose: Hand-pose estimation

As noted in [1], precise and high-resolution hand tracking enables a variety of AR/VR applications, including gaming, virtual education, teleoperation, haptics, embodiment realism, sports analytics, and rehabilitation in healthcare settings. Approaches to high-accuracy hand pose estimation include a wide range of techniques, such as monocular, multi-view, depth-based, motion capture, or infrared imaging with fixed or head-mounted cameras; wearable gloves equipped with magnetic, inertial, capacitive, or stretch sensors; smart rings; and wearables that utilize impedance tomography, inertial measurement units, acoustics, or ultrasound. However, all of these approaches come with their own limitations. Optical systems struggle in low-light environments and with occlusions. Multi-camera setups are more complex to install and calibrate. Glove-based wearables can limit dexterity, and forearm-mounted systems typically support only discrete gesture classification rather than full hand pose estimation [1].

Surface electromyography (sEMG) offers an alternative solution, as it is not limited by lighting conditions, visual occlusions, or the complex setup required by many other approaches. However, a key challenge with sEMG is learning a universal sEMG-to-pose decoder that generalizes to new participants and hand poses without requiring subject- or pose-specific training [1]. sEMG signals are inherently non-deterministic and exhibit domain shifts from device to device, user to user, and session to session with the same user due to factors such as electrode shift and muscle adaptation over time. Even within a single session, domain shifts can occur due to sweat, fatigue, and minor electrode displacements. According to Salter et al. [1], existing alternative datasets lack the scale necessary to address generalizability challenges in sEMG-based pose decoding. As a solution, they introduce the emg2pose dataset, which includes wrist sEMG recordings obtained using the sEMG-RD device and hand pose labels captured with motion markers tracked by a 28-camera setup. The dataset spans 193 users, 370 hours of data, and 29 different kinematic “stages” containing many low-level “gestures”, collected over 751 sessions. The dataset includes 80M labeled frames and is the largest of its kind [1].

Since sEMG signals are more closely related to movement than to static hand pose, the authors designed a decoding algorithm—vemg2pose (“Velocity-based emg2pose”)—to predict joint angular velocities. vemg2pose is autoregressive, meaning its prediction at time step t depends on the prediction at time step $t–1$. While we do not detail their model architecture here, its main building blocks include an embedding layer implemented via a causal strided convolutional featurizer, a Time-Depth Separable Convolution (TDS) network, and an LSTM

decoder, which outperformed a transformer-based model according to their experiments [1].

The dataset is divided into training and test stages, with the test stage further subdivided into three categories. The first subset contains unseen users performing in-distribution kinematic categories (referred to as "stages" in the original work). The second subset includes previously unseen kinematic categories (stages) but with in-distribution users. The third subset consists of both unseen users and unseen stages. They test their algorithm on these three test stages across two tasks: regression and tracking. In the regression task, the model must predict hand joint angles without any prior knowledge of the hand's position. In contrast, the tracking task assumes knowledge of the initial hand position, which makes predictions more accurate, particularly for early time steps where t is close to 0 [1].

In the regression setting, vemg2pose is compared against two previously validated hand pose decoding algorithms, NeuroPose [13] and SensingDynamics [14], with errors measured in terms of mean absolute joint angular error (°) and mean (Euclidean) landmark distance (mm), where landmarks correspond to joint and fingertip locations. Vemg2pose shows superior performance compared to the other two architectures.

Test Set	Baseline	Angular Error (°)	Landmark Distance (mm)
User	SensingDynamics	15.5 ± 1.4	21.8 ± 2.1
	NeuroPose	13.2 ± 1.1	17.5 ± 1.3
	vemg2pose	$\mathbf{12.2 \pm 1.3}$	$\mathbf{15.8 \pm 1.9}$
Stage	SensingDynamics	18.8 ± 1.6	26.6 ± 2.0
	NeuroPose	17.2 ± 1.7	24.0 ± 2.1
	vemg2pose	$\mathbf{15.2 \pm 1.6}$	$\mathbf{20.4 \pm 2.2}$
User, Stage	SensingDynamics	18.7 ± 1.6	27.2 ± 2.0
	NeuroPose	17.5 ± 1.5	24.9 ± 1.7
	vemg2pose	$\mathbf{15.8 \pm 1.4}$	$\mathbf{21.6 \pm 2.0}$

Table 1. Regression task: test set results comparing SensingDynamics, NeuroPose, and vemg2pose across user, stage, and user–stage test conditions. Mean ± standard deviations are reported across users. Bold indicates statistically significant improvement over baselines. Reproduced from [1].

Test Set	Baseline	Angular Error (°)	Landmark Distance (mm)
User	vemg2pose	7.7 ± 1.0	10.3 ± 1.5
Stage	vemg2pose	11.2 ± 1.4	15.2 ± 1.9
User, Stage	vemg2pose	11.0 ± 1.0	15.4 ± 1.4

Table 2. Tracking task: test set results for vemg2pose across user, stage, and user–stage test conditions. Mean ± standard deviations are reported across users. As can be seen, performance in the tracking task is higher than that of the regression task. Reproduced from [1].

Meta's contributions and shortcomings

The emg2pose dataset is a promising step toward cross-user, cross-gesture, and cross-session generalization for a variety of use cases, including neuroprosthetics, VR/AR applications, gaming, teleoperation, and other forms of human–computer interaction. Meta released similar datasets for different tasks like typed character classification [2], wrist angle estimation, pinch and thumb swipe movement classification, and handwriting character classification [3].

These datasets enable researchers to benchmark their decoding algorithms on datasets of unprecedented size within the field of wrist sEMG decoding. However, its utility for inference or real-world application is limited, as the dataset was collected using Meta's proprietary sEMG-RD device. Access is restricted to those who possess the hardware and are able to extract neural data from it. To the best of our knowledge, this is not possible with Meta's commercially available version of sEMG-RD. Furthermore, there has not been a study demonstrating independent use of the device by external researchers. In this paper, we propose an open-source alternative pipeline in which we design a similarly featured wrist-wearable sEMG signal acquisition device, outline an approach for obtaining labeled data, and fine-tune the pretrained vemg2pose model weights. This enables researchers to assemble the hardware, collect data, optimize the model through fine-tuning, and deploy the system in real-world, non-commercial applications since the original papers [1,2,3] are licensed under CC BY-NC-SA 4.0.

Our Contributions

For the reasons previously discussed, it is crucial that Meta's datasets be made usable for inference and real-world applications. This is especially important given that the emg2pose, emg2qwerty, and handwriting recognition datasets are the largest in their respective categories and address data collection challenges involving difficult labeling modalities [1,2,3]. In order to do so, one must have access to Meta's sEMG-RD or its commercial version and be able to collect neural data with it, which, to the best of our knowledge, has not yet been possible for external researchers. Alternatively, data can be collected using another sEMG device, and the appropriate algorithms can be fine-tuned. To make fine-tuning more effective, the alternative sEMG device should closely match the sEMG-RD in features, and a large amount of labeled data should be collected.

In this section, we highlight key technical considerations, review several existing and similar sEMG designs, present our proposed sEMG alternative, and describe a data collection pipeline for hand joint labeling to fine-tune vemg2pose.

General Considerations for sEMG Systems

The typical design of an sEMG acquisition system includes electrodes, differential preamplification, high-pass filtering, gain, detection, and low-pass filtering [5, p. 56]. These components must be selected to meet constraints on noise, bandwidth, impedance, and common-mode rejection.

sEMG signals occupy a bandwidth extending up to approximately 1 kHz, with most signal energy below 450–500 Hz [5]. Acquisition systems must therefore preserve frequency content at least up to 1 kHz to avoid signal distortion due to Nyquist-Shannon Theorem; however, in application scenarios, the Nyquist rate is taken to be 5 or 10 times instead of 2. Signal amplitudes range from tens of microvolts to several millivolts, with peak-to-peak values of up to approximately 5 mV during strong contractions [6].

According to Kearney [5], preamplifiers used in EMG systems should have moderate gains (0–100×) to avoid oversaturation while preserving signal fidelity. Overall system gain is typically in the range of 100 to 10,000 [7]. Input-referred noise should be less than 5 μV RMS to avoid drowning out the physiological signal, with 1–2 μV RMS often cited as a target for high-quality recordings [23].

The differential dynamic range of the system must be sufficient to amplify small EMG signals without distortion while tolerating baseline shifts and electrode-related DC offsets.

A high common-mode rejection ratio (CMRR) is critical for suppressing power-line interference. Differential preamplifiers used in EMG systems typically exhibit CMRR values of 100–120 dB at 60 Hz, with a reduction of 20–40 dB at 1 kHz considered normal [5]. Both SENIAM and ISEK guidelines recommend CMRR values exceeding 95 dB [24]. Battery operation further reduces coupling to power-line noise sources. Some designs use a notch filter at 50 or 60 Hz; however, this is usually not recommended.

The input impedance of the acquisition system must be much higher than the electrode-skin impedance to avoid signal attenuation. According to Kamen and Gabriel [6, p. 77], a common rule of thumb is that the input impedance should be at least 100 times greater than the source impedance. Skin impedance values exceeding 50 kΩ are reported as a worst-case scenario [24], highlighting the importance of high-impedance front-end design.

Current Alternatives to sEMG-RD

Table 3a: Comparison of comparable sEMG designs (Papers [1–18])

Paper	[1,2,3]	[15]	[16]	[17]	[18]
Device Name	sEMG-RD	α-armband v1	α-armband v2	HD-EMG	HD-EMG
# of Channels	16	16	16	32 Monop. Ch. per SU Module	32
Sampling Rate (kHz)	2	2	2	2.048	1
Electrodes	48x Gold-plated Brass 6.5 mm diameter	48x Gold-plated Copper 8 × 6 × 1 mm square	48x Gold-plated Copper 8 × 6 × 1 mm square	32x Silver	32x 50.27 mm^2 surface
Preamplification (Device or Gain)	190x	N/A	200x	192x	192x
Band-pass (Hz)	20–850	N/A–1000	2–1000	10–500	20–500
ADC - Resolution	12 bits	RHD2216–16 bits	16x AD8233–12 bits	RHD2132–16 bits	RHD2132–16 bits
Data Transfer	BLE	BLE	BLE	Wi-Fi	2.4 GHz ISM Wireless Link
Microcontroller	N/A	STM32F765VIH6	STM32F765VIH6	CC3200	MSP430F5529
Input-referred Noise (μV RMS)	2	2.4	N/A	1.8 ± 0.2	2.4
Input Impedance (AFE or ADC)	10pF ‖ 10TΩ	N/A	N/A	1GΩ at 50 Hz	N/A
CMRR (dB)	N/A	N/A	N/A	82 dB	N/A
ADC Dynamic Range	65.5 dB	N/A	N/A	±5 mVpp	±5 mVpp
Max Input V	6.6	N/A	N/A	N/A	N/A
Reference / RLD Electrode(s)	No	1 Ref. Elec. per Ch.	1 Ref. Elec. per Ch.	1 Ref. Elec.	16 Ref. Elec.
Open-source	No	No	Not Fully**	No	No
Price	$799, $199*	N/A	$150	N/A	N/A

Table 3b: Comparison of comparable sEMG designs (Papers [19–22])

Paper	[19]	[20,21]	[22]
Device Name	WyoFlex	BioWolf16	Circular Wireless sEMG
# of Channels	4	16	8
Sampling Rate (kHz)	1.6	up to 4	1
Electrodes	4x Metal	16x Gold-plated demagnetized neodymium or 16x Ag/AgCl	36103166S by Wurth Elektronic 16.1 × 16.1 × 3.6 mm
Preamplification (Device or Gain)	N/A	AD8603–1-12x	INA333
Band-pass (Hz)	0.1–500	Digital HP at 2 Hz–No LP	20–2****
ADC - Resolution	Gravity Analog EMG sensor–12 bits	2xADS1298–24 bits	N/A
Data Transfer	Wi-Fi	BLE	BLE
Microcontroller	FireBeetle ESP32-E	Mr Wolf	Nordic nRF52840
Input-referred Noise (μV RMS)	N/A	1.62 µV RMS at PGA=1	N/A
Input Impedance (AFE or ADC)	N/A	N/A	N/A
CMRR (dB)	N/A	115 dB to 122 dB at 50 Hz	100 dB at 50 Hz
ADC Dynamic Range	±1.5 mV.	N/A	N/A
Max Input V	3	N/A	N/A
Reference / RLD Electrode(s)	N/A	Ref. and Grn. Elec. on Elbow	1 Ref. Elec.
Open-source	No	Bracelet design	Electronics and mechanical designs
Price	$250	EUR30 for mass production***	EUR100

*Meta's commercial bracelet can be purchased for $199 on its own or bundled with their Ray-Ban smart glasses for $799, but it does not provide access to neural data.
**The authors claim, "Any interested scholar or engineer may contact the corresponding author to promptly obtain materials such as hardware circuit project files, software code, and algorithm models." [16] However, we have not received any replies to our email requesting access to the files. Even if we had, the files are not readily accessible as open-source.
***The prices may not be reflective of now, as the two ADS1298 chips utilized in BioWolf16 itself costs 35 Euros for large orders.
****The article claims Low-Pass Filter was set at 2 Hz; however, it is most likely referring to 2 kHz. " The last part of the circuit is an active lowpass filter with adjustable gain. The cutoff frequency of this filter is equal to 2 Hz." [22]

In Table 3, we provide a high-level comparison of recent dry electrode sEMG designs, which range from 4 to 32 channels and have sampling rates between 1 kHz and 4 kHz [1,2,3,15–22]. The devices listed represent a subset of our literature review, selected for having features similar to those of the sEMG-RD. The closest match is the α-armband v2. However, this armband is not open source and uses a single ADC for every three electrodes, resulting in a total of 16 ADCs [16]. We believe this design can be further optimized and simplified by reducing the number of ADCs, and that it should be made open-source so the public can readily implement it.

Another close match is the armband based on the BioWolf16 [20,21], which closely resembles our proposed design. However, it uses demagnetized gold-plated neodymium, and the demagnetization process typically requires a degausser, which is not readily accessible to most users. Furthermore, their reported cost of 30 Euros appears inaccurate, as the ADCs alone cost approximately 35 Euros at current exchange rates. As an alternative, we propose using gold-plated brass pogo pins,

which are more accessible and provide an accurate imitation of the sEMG-RD design. We also eliminate the use of front-end amplification to simplify the circuitry.

System Overview and Design Philosophy

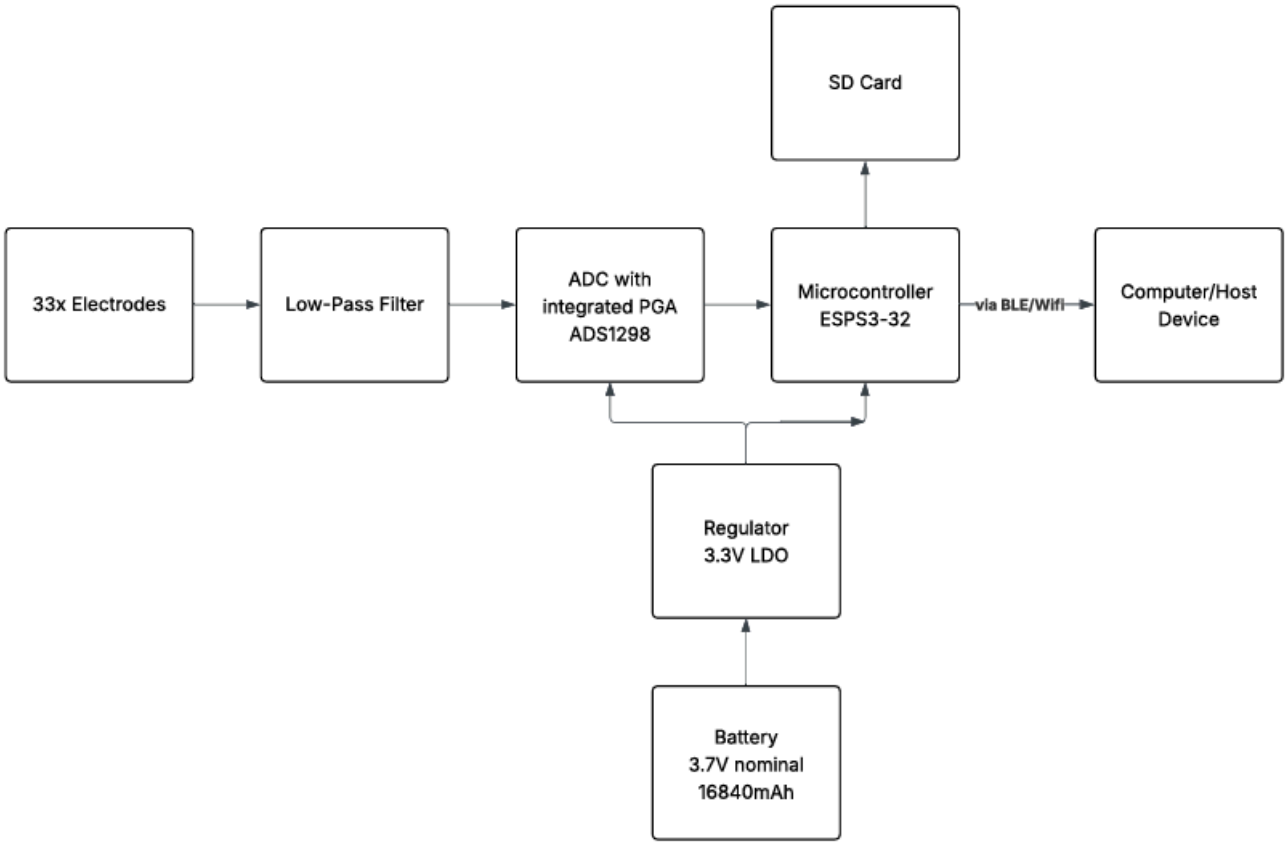


Figure 2: Our proposal's high-level schematic.

As mentioned in the Electromyography" section, the signals that we intend to pick up with sEMG are on the magnitude of a couple μ*V* to a +-5 mVpp [4,6]. Because we are working with such small signals, it is important to minimize noise and prevent interference from the circuitry itself.

In this subsection, we propose a wrist-worn, multi-channel sEMG acquisition system designed to capture low-amplitude biopotential signals and transmit them wirelessly for downstream processing and joint-angle estimation.

The system consists of surface electrodes, a multi-channel biopotential analog front-end with integrated amplification and analog-to-digital conversion, a

microcontroller unit (MCU) for data acquisition and wireless transmission, and a software pipeline for preprocessing and learning-based inference.

To satisfy the requirements outlined in "General Requirements for sEMG systems" subsection, we selected the Texas Instruments ADS1298, a low-power, 8-channel, 24-bit biopotential analog front-end with integrated programmable gain amplifiers and delta-sigma ADCs [6]. The ADS1298 is specifically designed for electrophysiological measurements and integrates features that would otherwise require multiple discrete components.

Rather than implementing a traditional discrete analog front-end with separate instrumentation amplifiers, low-pass filters, and gain stages, we deliberately simplify the analog circuitry. Our design relies primarily on the integrated programmable gain amplifier (PGA), delta-sigma ADC, and digital decimation filtering capabilities of a dedicated biopotential analog front-end. Analog filtering is limited to anti-aliasing, with DC offset and baseline drift handled digitally where feasible. Our design choice reduces circuit complexity and improves reproducibility while still satisfying established sEMG performance requirements.

Input Impedance

The ADS1298 offers selectable input impedance modes. In the configuration used for this design, the input impedance is approximately 500 MΩ to 1000 MΩ, depending on lead-off detection settings [6]. Using the 500 MΩ configuration allows accommodation of electrode-skin impedances on the order of several megaohms, well beyond realistic worst-case values.

Common-Mode Rejection Ratio

At a PGA gain of 12, the ADS1298 provides a CMRR exceeding 115 dB at 60 Hz and greater than 90 dB at 1 kHz, satisfying SENIAM and ISEK recommendations [25, p. 18].

Input-referred Noise

When operated at sampling rates of 2 ksps or 4 ksps in either low-power or high-resolution modes with a PGA gain of 12, the expected input-referred noise of the ADS1298 is approximately 0.9 to 1.5 μV RMS, corresponding to roughly 6 to 11.3 μV peak-to-peak [25]. This performance is well within the noise limits recommended for high-quality sEMG acquisition.

Dynamic Range and Resolution

We plan to use an internal reference voltage of 2.4 V. With a PGA gain of 12, this yields a full-scale differential input range of approximately 200 mV [6, p. 32]. Given that the expected maximum sEMG amplitude is approximately 5 mV, only about 2.5% of the available full-scale range is utilized by meaningful EMG signals. However, the resulting voltage resolution remains sufficiently small and having a larger full-scale range gives us some protection against DC offset and movement artifact oversaturating our ADC, as we don't have an analog HP filter:
$resolution\ in\ volts\ =\ (VRef\ /\ Gain)\ /(\ 2^{23}-1\ )\ =\ 0.023\ microvolts,$
$given\ VRef\ =\ 2.4V\ .\ Gain\ =\ 12$

Analog Front-End and Filtering

Most EMG systems employ DC-coupled differential preamplifiers to preserve CMRR, as introducing capacitors between electrodes and amplifiers can degrade CMRR due to component tolerances. Even capacitors with 1% tolerance introduce a mismatch that negatively affects common-mode rejection.

In this design, we avoid analog high-pass filtering and instead address DC offset and baseline drift digitally. This approach simplifies the circuit and preserves CMRR while relying on the ADC's high resolution to maintain signal fidelity. However, we must ensure that our DC offset + Movement Artifact + Muscle Signal does not exceed our full-scale voltage (200mV). We need to test our current gold-plated brass electrodes from Mill-Max Manufacturing with 2.24mm diameters in order to make sure saturation is not likely. If unsuccessful alternatives like [26] will be considered.

Each channel of the ADS1298 includes a 24-bit delta-sigma ADC with a second-order modulator optimized for low-power operation [25, p. 32]. In high-resolution mode, the modulator sampling frequency is fMOD = fCLK / 4 and in low-power mode fMOD = fCLK / 8, where fCLK=2.048 MHz. As with all delta-sigma converters, quantization noise is shaped to frequencies up to fMOD/2.

The ADS1298 includes on-chip digital decimation filters that provide anti-aliasing up to fMOD/2, significantly reducing the complexity of external analog filtering required compared to Nyquist ADCs [25, p. 32].

External RC Anti-Aliasing Filter and CMRR Impact

The most important benefit of having a high CMRR comes from being able to eliminate 60Hz power-line noise. To further eliminate this noise, our design is powered by a battery instead of being plugged to a power outlet, which also helps with electrical safety.

Our ADC has a CMRR of more than 90 dB even at 1kHz with Gain=12, and >115 dB at power-line hum frequency (60 Hz for North America)

To attenuate interference near multiples of fMOD, a simple RC low-pass filter is used at the input [6, p. 49]. For example, using 1 kΩ resistors with 0.1% tolerance and 620 pF capacitors with 1% tolerance yields an anti-aliasing cutoff near 128kHz. Under these conditions, the resulting CMRR is approximately 105 dB at 60 Hz and 81 dB at 1 kHz. These values improve to approximately 111 dB and 87 dB, respectively, when the cutoff is increased to 256 kHz [25, p. 88]. The equation to calculate this is as follows: where fc is the – 3-dB frequency of the RC filter.

$$CMR = 20log(\delta R/R + \delta C/C) + 20log(f/fc),$$

For LP filtering at 128 kHz with 0.001 tolerance resistors and 0.01 tolerance capacitors, the CMR is as follows:

$$CMR\ at\ 60\ Hz = 20log(0.001 + 0.01) + 20log(60/128000) = 105dB$$

$$CMR\ at\ 1\ kHz = 20log(0.001 + 0.01) + 20log(1000/128000) = 81dB$$

For LP filtering at 256kHz with 0.001 tolerance resistors,and 0.01 tolerance capacitors, the CMR is as follows:

$$CMR\ at\ 60\ Hz = 20log(0.001 + 0.01) + 20log(60/256000) = 111dB$$

$$CMR\ at\ 1\ kHz = 20log(0.001 + 0.01) + 20log(1000/256000) = 87dB$$

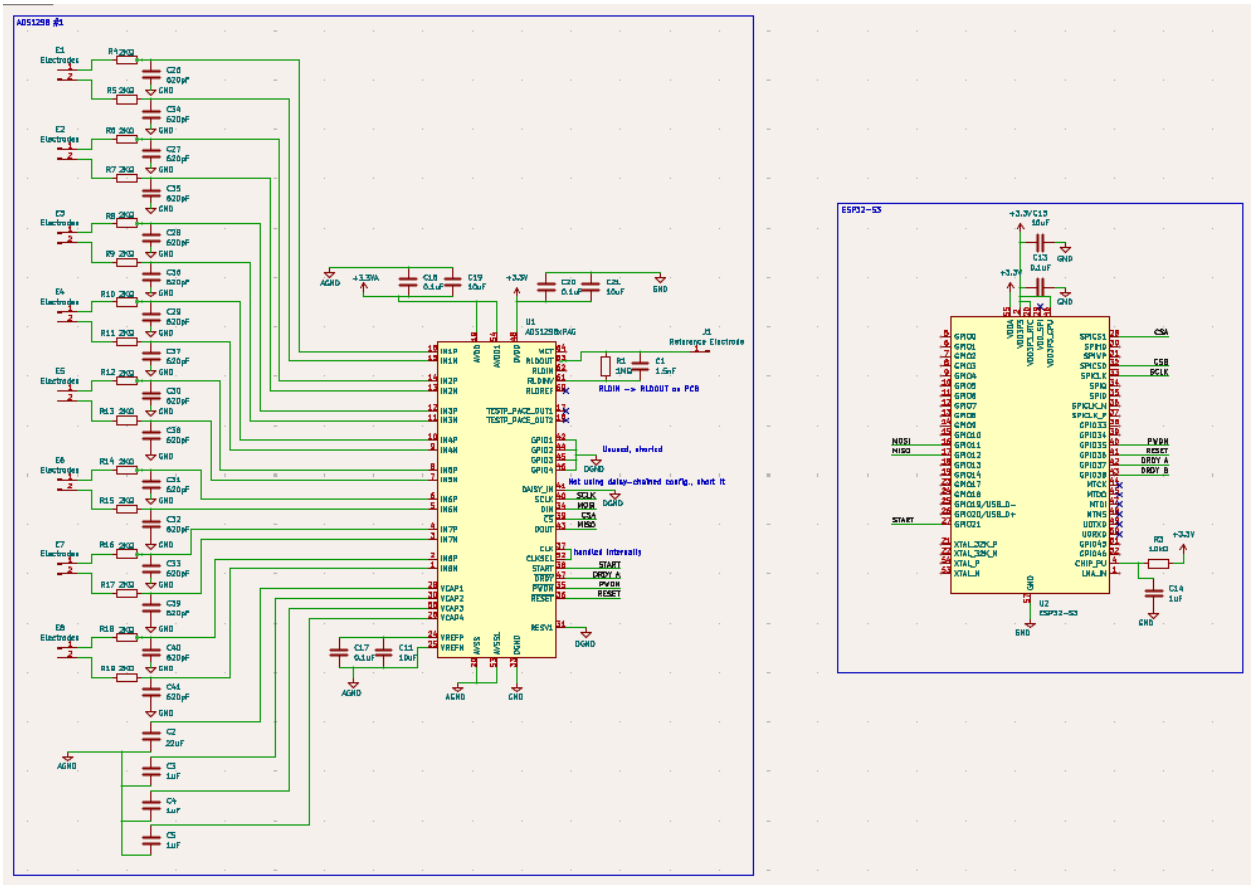


Figure 2: Schematics of Electrode - AFE (RC Low-Pass) - ADC - MCU for one ADC only due to size limitations. The files will be available online after the power regulation and the daisy-chain configurations are added.

DC Offset and Baseline Drift

ADC offset error is approximately 500 µV, with gain error around ±0.2% of full-scale and integral nonlinearity on the order of 6 ppm at the operating point [25]. DC offset will be addressed through digital high-pass filtering or baseline correction at the MCU level. The right-leg drive (RLD) that will be placed on the Ulna bone is also expected to help control common-mode voltage and DC offset, though its exact contribution will be validated experimentally.

Given the full-scale differential range of approximately 200 mV, saturation would require DC offset values approaching 150 mV [25, p. 12], which have been reported when using carbonized rubber electrodes [27]. However, to the best of our knowledge, there are no reports of DC offsets approaching these values when using gold-plated electrodes.

We will use a digital High-Pass filter to remove the DC offset and low-frequency movement artifacts, the cutoff frequency will be set to 20 Hz.

RLD will be implemented following TI application guidelines to further improve common-mode rejection [27]. Additional noise reduction strategies include short cable lengths, shielding, and careful PCB layout. Breadboard-based testing is expected to be noisy and is used only for functional validation.

Data Rate and Throughput

Sigma–Delta ADCs, such as the ADS1298, perform oversampling, and their output rate corresponds to the amount of data they transmit, effectively acting as the sampling rate. Accordingly, our final output rate will be 2 kHz, which will attenuate signals above 524 Hz, as shown in the table below. The data will be transmitted to our microcontroller (MCU), which will then send it wirelessly to a host computer. In parallel, the data will also be saved locally on an SD card to ensure recovery in case of wireless transmission failure, allowing the data to remain usable for unsupervised and self-supervised training regimes.

Table 1. Input-Referred Noise μV_{RMS} (μV_{PP}) in High-Resolution Mode 3-V Analog Supply and 2.4-V Reference(1)

DR BITS OF CONFIG1 REGISTER	OUTPUT DATA RATE (SPS)	–3-dB BANDWIDTH (Hz)	PGA GAIN = 1	PGA GAIN = 2	PGA GAIN = 3	PGA GAIN = 4	PGA GAIN = 6	PGA GAIN = 8	PGA GAIN = 12
000	32000	8398	335 (3553)	168 (1701)	112 (1100)	85 (823)	58 (529)	42.5 (378)	28.6 (248)
001	16000	4193	56 (613)	28 (295)	18.8 (188)	14.3 (143)	9.7 (94)	7.4 (69)	5.2 (44.3)
010	8000	2096	12.4 (111)	6.5 (54)	4.5 (37.9)	3.5 (29.7)	2.6 (21.7)	2.2 (17.8)	1.8 (13.8)
011	4000	1048	6.1 (44.8)	3.2 (23.3)	2.4 (17.1)	1.9 (14)	1.5 (11.1)	1.3 (9.7)	1.2 (8.5)
100	2000	524	4.1 (27.8)	2.2 (15.4)	1.6 (11)	1.3 (9.1)	1.1 (7.3)	1 (6.5)	0.9 (6)
101	1000	262	2.9 (19)	1.6 (10.1)	1.2 (7.5)	1 (6.2)	0.8 (5)	0.7 (4.6)	0.6 (4.1)
110	500	131	2.1 (12.5)	1.1 (6.8)	0.9 (5.1)	0.7 (4.3)	0.6 (3.5)	0.5 (3.1)	0.5 (2.9)

(1) At least 1000 consecutive readings used to calculate the RMS and peak-to-peak noise values in this table.

Table 2. Input-Referred Noise μV_{RMS} (μV_{PP}) in Low-Power Mode 3-V Analog Supply and 2.4-V Reference(1)

DR BITS OF CONFIG1 REGISTER	OUTPUT DATA RATE (SPS)	–3-dB BANDWIDTH (Hz)	PGA GAIN = 1	PGA GAIN = 2	PGA GAIN = 3	PGA GAIN = 4	PGA GAIN = 6	PGA GAIN = 8	PGA GAIN = 12
000	16000	4193	333 (3481)	166 (1836)	111 (1168)	84 (834)	56 (576)	42 (450)	28 (284)
001	8000	2096	56 (554)	28 (272)	19 (177)	14.3 (133)	9.7 (85)	7.4 (64)	5 (42.4)
010	4000	1048	12.5 (99)	6.5 (51)	4.5 (35)	3.4 (25.9)	2.4 (18.8)	2 (14.5)	1.5 (11.3)
011	2000	524	6.1 (41.8)	3.2 (22.2)	2.3 (15.9)	1.8 (12.1)	1.4 (9.3)	1.2 (7.8)	1 (6.7)
100	1000	262	4.1 (26.3)	2.2 (14.6)	1.6 (9.9)	1.3 (8.1)	1 (6.2)	0.8 (5.4)	0.7 (4.7)
101	500	131	3 (17.9)	1.6 (9.8)	1.1 (6.8)	0.9 (5.7)	0.7 (4.2)	0.6 (3.6)	0.5 (3.4)
110	250	65	2.1 (11.9)	1.1 (6.3)	0.8 (4.6)	0.7 (4)	0.5 (3)	0.5 (2.6)	0.4 (2.4)

(1) At least 1000 consecutive readings used to calculate the RMS and peak-to-peak noise values in this table.

Figure 4: Adapted from [25]

Streaming 16 channels at 2 kHz with 24-bit samples requires 16 × 2000 × 24 = 768,000 bit/s (768 kbps).After accounting for framing and minimal packet headers, the practical requirement is approximately 0.85 to 0.90 Mbps.BLE on the ESP32-S3 provides a maximum throughput of about 700 kbps under favorable conditions, which is below the required data rate. In contrast, Wi-Fi on the ESP32-S3 supports more than 20 Mbps of usable throughput, which comfortably exceeds the sub-1 Mbps requirement. Therefore, Wi-Fi is the appropriate choice for reliable real-time streaming, as BLE is not sufficient for full-rate 16-channel, 2 kHz, 24-bit sEMG data [29].

Device Safety

Our proposed sEMG device is powered exclusively by a 3.7 V LiPo battery during operation and is explicitly not intended to be worn while charging or connected to any wired interface. This design choice eliminates any direct mains-referenced electrical path to the user and substantially reduces the risk of electric shock compared to line-powered systems. Residual safety considerations are therefore limited to battery-related hazards and applied-part currents. Battery risks are mitigated through the use of a protected cell and appropriate charging circuitry operated only off-body, as well as mechanical enclosure design that prevents puncture or short-circuiting. As a result, the system's safety profile is dominated by low-energy, battery-limited operation, making it suitable for use as a wearable sEMG acquisition device when operated according to the stated constraints.

Data Collection

For data collection, we will utilize Meta's recently announced SAM3DB (Body) model [30]. SAM3DB operates by identifying the body parts present in an image and fitting a 3D human mesh, called the Momentum Human Rig [31], onto 2D images. This mesh incorporates joints, which enable the calculation of joint angles. We have observed that the model performs particularly well when the upper body of the person is visible in the frame or when the point of view is third-person. In contrast, images collected from head-mounted cameras in a first-person perspective are unlikely to yield accurate body mappings.

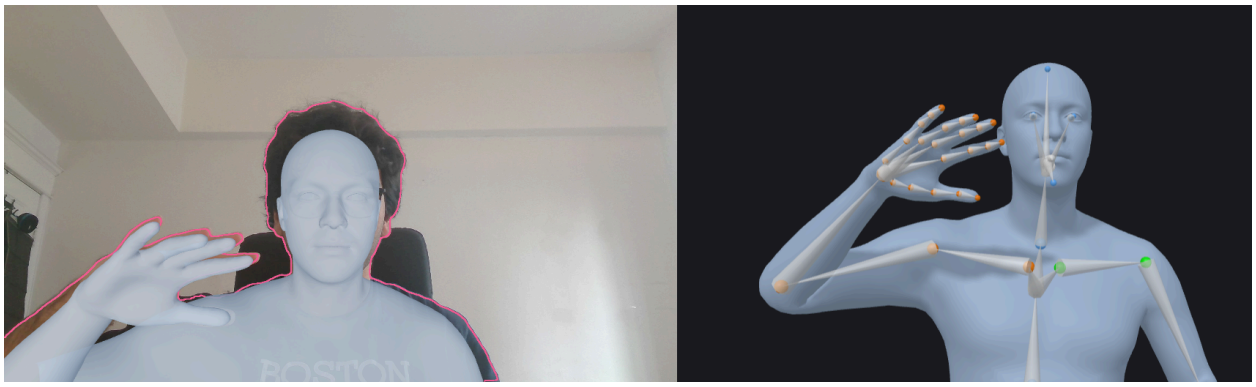


Figure 5: The model performs accurately in a frame where most of the upper body is present

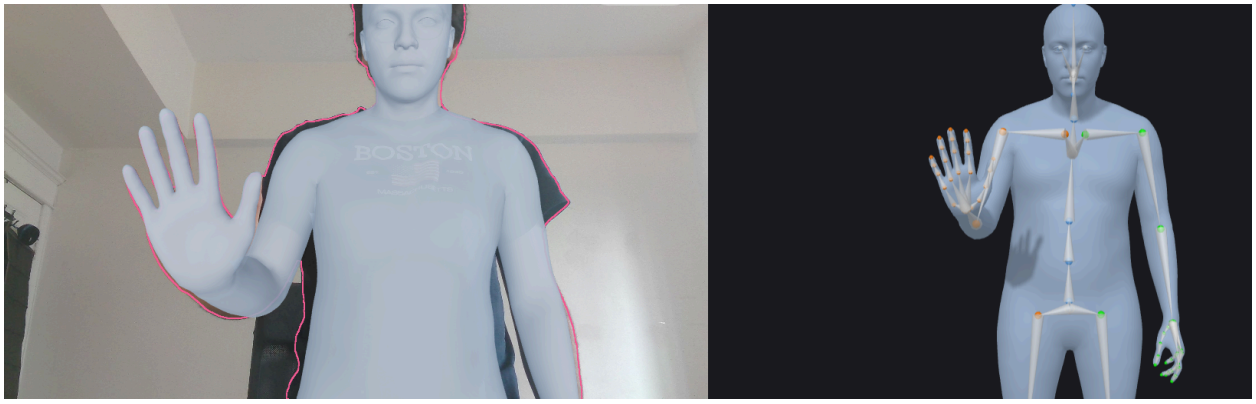


Figure 6: The model performs accurately in a frame where the upper body is present

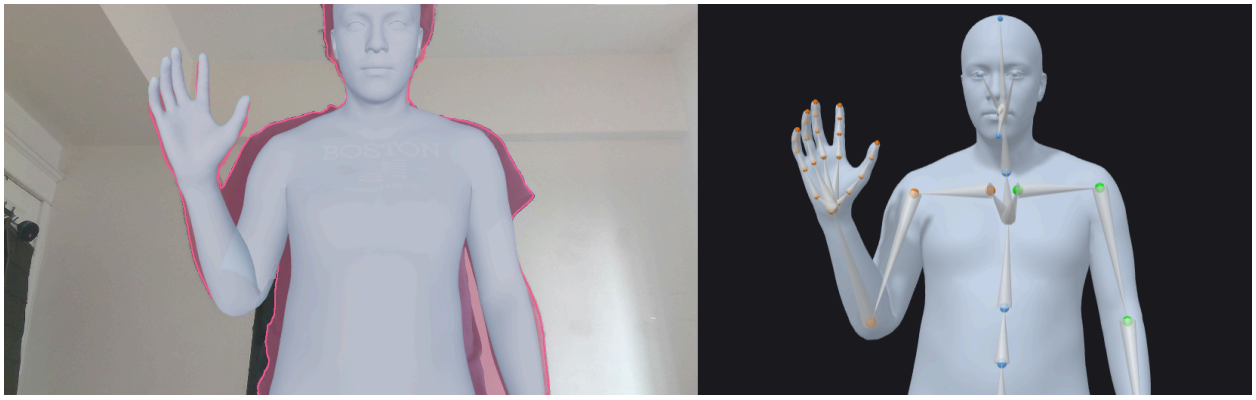


Figure 7: The model performs accurately in a frame where the upper body is present

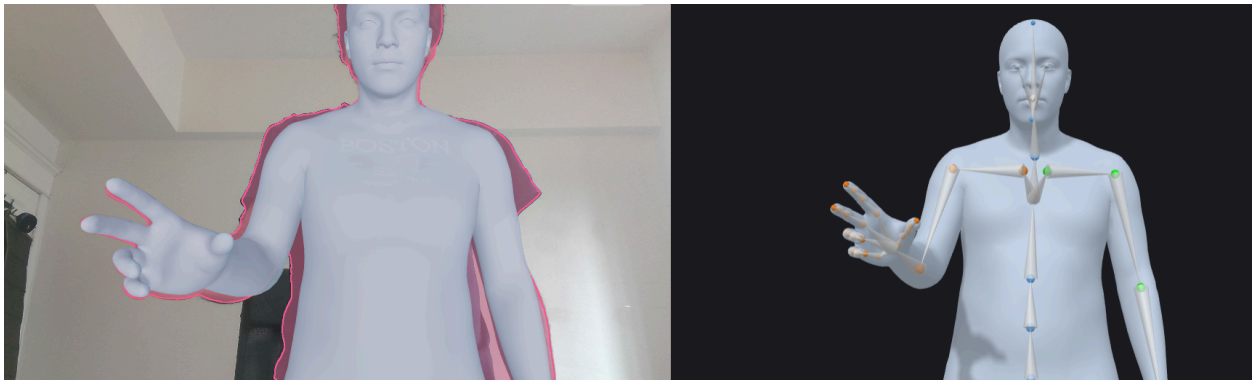


Figure 8: The model performs accurately in a frame with some occlusions while the upper body is present

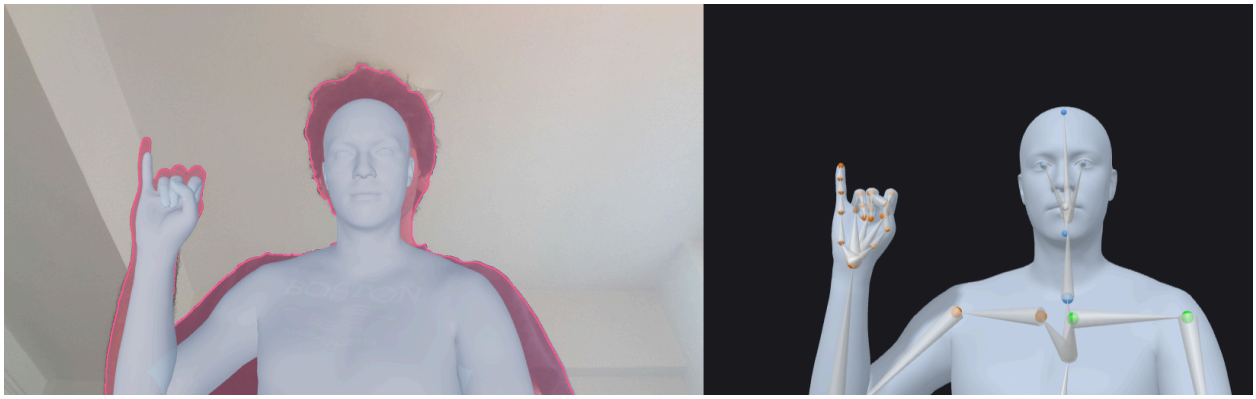


Figure 9: The model performs accurately in a frame with heavy occlusions while the upper body is present

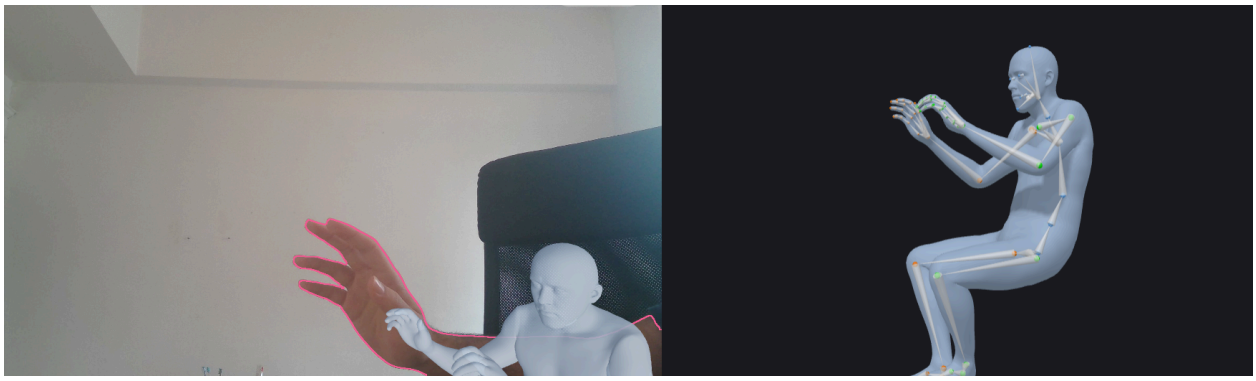


Figure 10: The model performs very poorly where the upper body is not present

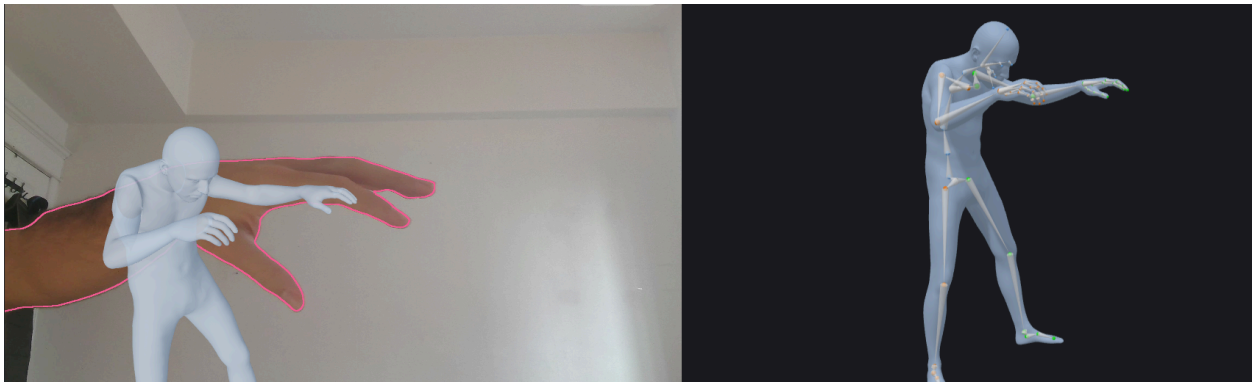


Figure 11: The model performs very poorly where the upper body is not present

As shown in Figures 5 through 11, the model struggles to determine body position in frames that do not include the subject's upper body. Therefore, we will position our camera to ensure a clear view of the subjects' upper bodies and upper limbs during the data collection process.

The remaining potential pitfalls of our data collection process are data synchronization and SAM3DB's compute requirements. For compute requirements, SAM3DB takes around 20-30 seconds per frame on mid-grade consumer GPUs like RTX 3070 and more time on CPUs; therefore is unable to live inference in most cases. We will utilize Nebula Block's H100 cluster; however, those following this paper to collect data must find other ways to access compute.

To achieve accurate temporal alignment, i.e. data synchronization, between sEMG recordings and video frames, we will employ a hybrid synchronization strategy that combines hardware-level markers with timestamp-based alignment. At the beginning of each recording, and periodically throughout acquisition, the microcontroller emits a synchronization pulse. This pulse is recorded directly in the sEMG data stream under the same sampling clock as the electrophysiological signals and is simultaneously rendered in the video stream via an LED placed within the camera's field of view. These hardware markers provide unambiguous anchor points for correcting the initial temporal offset and for estimating relative clock drift between the sEMG acquisition system and the video capture device.

In parallel, all sEMG packets are going to be timestamped at the source using the microcontroller's local clock, while video frames are indexed using capture-time metadata on the host system. Timestamp alignment enables a continuous temporal mapping between the two modalities, while the hardware markers serve to validate this mapping and correct for accumulated drift or transport-layer jitter introduced

by wireless transmission. By combining robust physical synchronization cues with flexible software-level timestamps, this approach enables reliable sub-frame alignment over long recording durations without requiring shared clocks or external trigger hardware.

Conclusion & Discussion

This project presents a practical and reproducible pathway toward democratizing high-performance wrist sEMG research by decoupling state-of-the-art decoding algorithms from proprietary hardware. We designed a fully open-source, wrist-worn sEMG acquisition system that closely matches the electrical, temporal, and spatial characteristics of Meta's sEMG-RD, while deliberately simplifying the analog front end to improve accessibility, robustness, and reproducibility. By utilizing a modern biopotential ADC with integrated AFE (ADS1298), battery-powered operation, high input impedance, high CMRR, and digitally handled baseline correction, the proposed design satisfies established sEMG requirements while remaining feasible for independent researchers to build, modify, and deploy

Beyond hardware, we outlined a complete end-to-end pipeline for enabling inference and adaptation on Meta's large-scale datasets, with particular emphasis on emg2pose. We identified the central barrier to real-world use of these datasets, namely the lack of accessible hardware for data acquisition, and proposed a concrete solution that enables fine-tuning of pretrained vemg2pose models using newly collected data. The proposed data collection strategy, combining camera-based 3D body reconstruction with a hybrid hardware–software synchronization scheme, provides a scalable alternative to large motion-capture studios while preserving temporal alignment accuracy required for supervised learning.

The remaining work is straightforward but essential: validate DC offset and motion artifact behavior of the specific gold-plated brass electrodes we are using, confirm that saturation risk is low in practice given our full-scale range, benchmark wireless reliability at full rate, and quantify how much fine-tuning data is needed to recover performance relative to the original sEMG-RD setup. If those pieces check out, this project turns Meta's datasets from "benchmark-only" into something that can actually be deployed and extended by the broader research community.

Future Work

Several components of the proposed system remain to be validated and completed. First, the impedance characteristics and average DC offset of the gold-plated brass electrodes must be experimentally measured, as these properties are not well documented in the existing literature. Second, the hardware design needs to be finalized by incorporating power-management circuitry, including the battery, charging protection, and low-dropout regulators, followed by PCB layout and fabrication. On the mechanical side, the system must be integrated into a wearable form factor, potentially by adapting the bracelet design described in [20]. From a systems perspective, end-to-end communication must be tested and validated, spanning the signal path from the electrodes to the ADC, from the ADC to the microcontroller, and from the microcontroller to the host device. Finally, the SAM3D-based automated data collection pipeline must be completed, with particular emphasis on dependable timestamping between the video frames and the sEMG recordings.

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